Yes: the basics are in Step 1 (realized volatility) and Step 3 (the model). Here they are in one place:

- **What it forecasts:** tomorrow's volatility (how much price will move), not the direction.
- **What it uses:** realized volatility (RV), one number per day made by adding up the squared 5-minute returns of that day.
- **The idea:** tomorrow's RV = constant + b_1 × yesterday's RV + b_2 × last 5 days' average RV + b_3 × last 22 days' average RV.
- **Why three averages:** short-term, medium-term and long-term traders each push volatility; together the three averages copy volatility's long memory.
- **How it is fitted:** plain OLS regression, the same as in your econometrics course. No optimiser, nothing that can fail to converge.
- **How it is used:** $\sqrt{}$(forecast RV) = expected daily move in %, for stops and position size.

HAR-RV: A Simple Lesson

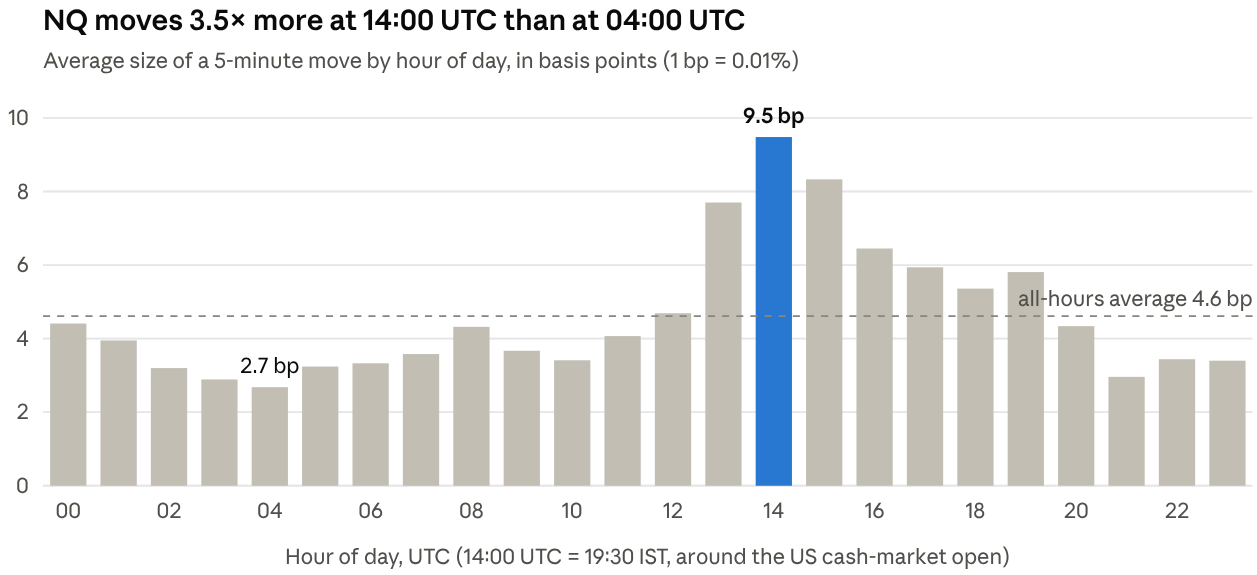
Oct 17, 2026 · @anandpc

Full form: HAR-RV = Heterogeneous AutoRegressive model of Realized Volatility. "Heterogeneous" = traders with different time horizons (day, week, month); "AutoRegressive" = it predicts from its own past values; "Realized Volatility" = the daily volatility measured from intraday returns.

HAR-RV forecasts tomorrow's volatility as a weighted mix of three averages of realized volatility: the last day, the last week and the last month. It is a plain linear regression, so if you know OLS from your econometrics course, you already know how to fit it. On one year of your NQ 1-minute data it beat daily GARCH(1,1) on 70 days it had never seen (R^2 0.39 against 0.20).

Why HAR-RV exists

Volatility follows the clock, and GARCH(1,1) does not know the clock.



your differenced_data.csv · NQ 1-minute data, 29 Sep 2025 – 29 Sep 2026, grouped into 5-minute moves

GARCH(1,1) works bar by bar and treats every bar the same, so it mistakes this daily rhythm for news: it raises volatility at every US open and lowers it every Asian night. HAR-RV sidesteps the problem by first adding a whole day of small moves into one number, realized volatility, and then forecasting that number. The daily rhythm is inside every day, so it no longer confuses the model.

Step 1: Realized volatility

Realized volatility (RV) measures one day's volatility by adding up that day's small moves, instead of looking at one close-to-close return.

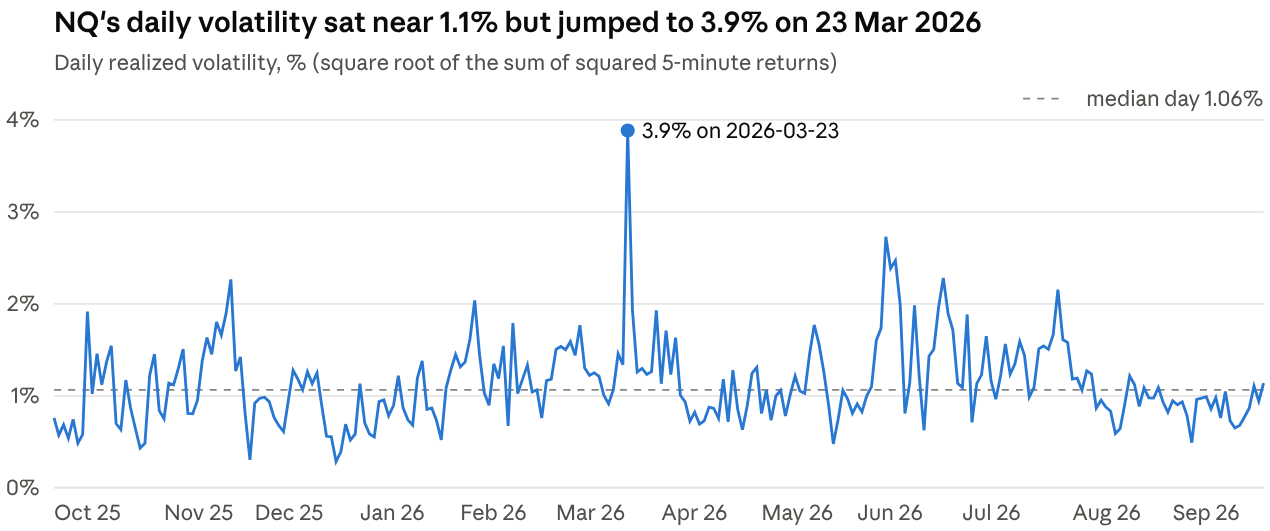
```
RV_t = \sum_{i=1}^{M} r_{t,i}^{2}
```

Here r(t,i) is the i-th 5-minute log return on day t, and M is the number of 5-minute bars that day (about 270 for NQ). RV is a variance; its square root is the day's realized volatility, in % when the returns are in %.

A small example. Suppose a day had only four 5-minute moves: +0.2%, −0.1%, +0.3% and −0.2%. Then RV = 0.04 + 0.01 + 0.09 + 0.04 = 0.18 %², and the day's volatility = $\sqrt{0.18} \approx 0.42\%$.

Why 5 minutes and not every tick. At tick level the bid-ask bounce adds noise that makes RV too big. About 5 minutes is the usual compromise between using lots of data and avoiding that noise.

Why it beats one daily return. A single close-to-close return can be near zero on a day that swung wildly and came back. RV sees every swing, so it is a far sharper measure of how volatile the day really was.



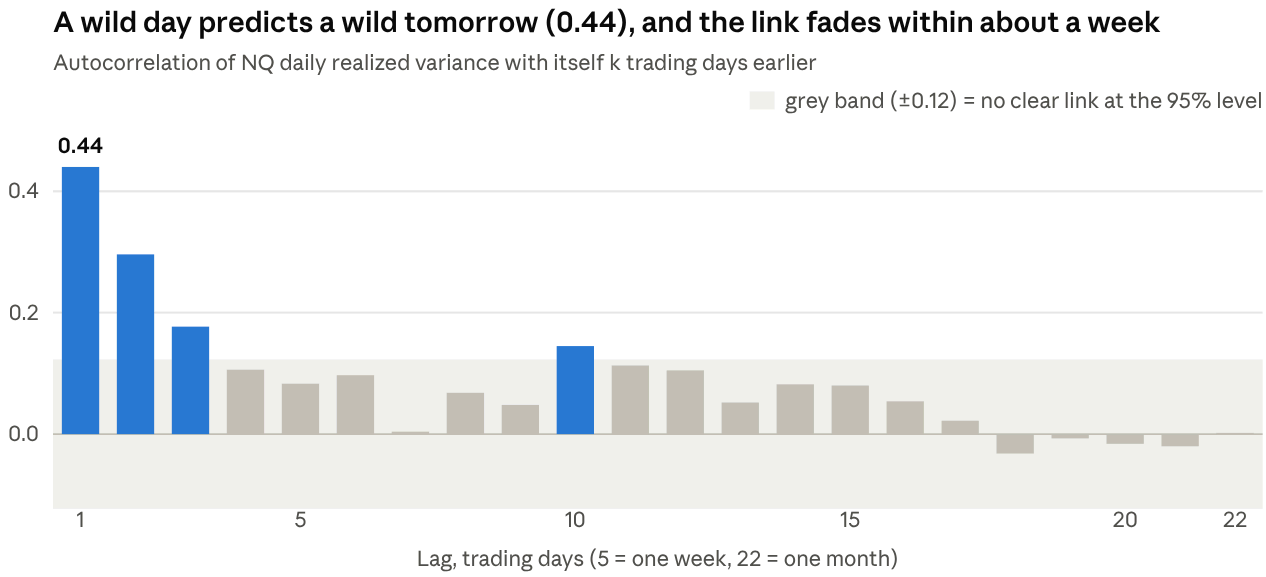
your differenced_data.csv · 254 full trading days, 1 Oct 2025 – 28 Sep 2026

This line is the series HAR-RV forecasts. Calm stretches and wild stretches each last several days, which is what makes it forecastable.

Step 2: Volatility has memory

A wild day is usually followed by another wild day, so past volatility helps predict future volatility.

The graph below is an ACF (autocorrelation function), not a PACF. Each bar is the plain correlation between today's RV and the RV k days earlier, including any effect passed through the days in between. A PACF would remove those in-between effects and show only the direct link at each lag. HAR-RV uses the ACF view because its weekly and monthly averages are designed to capture exactly this slow, cumulative fade.



your differenced_data.csv · 254 trading days of NQ realized variance, lags 1–22

How to read it. Each bar is a correlation: how strongly today's RV moves with RV k days earlier. A tall bar at lag 1 means yesterday tells you a lot about today; bars inside the grey band are too small to trust. The blue bars are the lags where the link is clearly real.

What it shows. The link is strong for the first few days, shrinks through the week, and has a smaller bump around two weeks. On many years of stock-index data this memory lasts for months; one year of NQ shows a shorter version.

Why it matters for the model. A memory that is strong at first and then fades slowly is hard for one short-memory term to copy. HAR-RV copies it by mixing a 1-day, a 5-day and a 22-day average.

Step 3: The HAR-RV model

The whole model is one line: tomorrow's realized variance = a constant + a weight on yesterday + a weight on last week's average + a weight on last month's average.

```
RV_{t+1} = \beta_0 + \beta_d \, RV_t + \beta_w \, RV^{(w)}_t + \beta_m \,
RV^{(m)}_t + \varepsilon_{t+1}
```

```
RV^{(w)}_t = \frac{1}{5}\sum_{i=0}^{4} RV_{t-i} \qquad RV^{(m)}_t = \frac{1}
{22}\sum_{i=0}^{21} RV_{t-i}
```

Part	Window	Who it stands for	Plain meaning
Daily, RV_t	1 day	day traders, scalpers	what happened yesterday
Weekly, RV(w)	last 5 trading days	swing traders	the mood of the past week
Monthly, RV(m)	last 22 trading days	funds, long-term investors	the background level

Why the name. HAR = Heterogeneous AutoRegressive. "Autoregressive" because it predicts volatility from its own past, like an AR model. "Heterogeneous" because markets hold traders with different time horizons, and each average stands for one group.

Why it works so well with so little. Three overlapping averages fade slowly when added together. That lets four numbers copy the long, slow memory of volatility that ARMA and GARCH(1,1) struggle to capture.

Why it is easy. It is ordinary least squares (OLS). There is no optimiser that can fail to converge, it fits in milliseconds, and every regression tool from your course applies: t-stats, R^2, residual checks.

Step 4: Fitting it

On your NQ data, yesterday's volatility carried almost all the weight: 0.36, the only clearly significant term. The model was fitted by OLS on 162 trading days (1 Oct 2025 to 19 Jun 2026), with RV in $\%^2$ per day.

Term	Coefficient	t-stat	What it means
Constant	0.82	3.32	the baseline every forecast starts from
Daily	0.36	2.68	36% of yesterday's variance carries into tomorrow
Weekly	0.12	1.27	weak; not significant in this sample
Monthly	−0.01	−0.04	no effect in this sample

Reading the weights. The three slopes add up to 0.47. The rest (0.53) is the pull back to the long-run level, which the model puts at 0.82 / 0.53 = 1.55 %², or about **1.25% a day**. The average day in the data was 1.13%, so the model expects volatility to drift back towards that range after a calm or a wild spell.

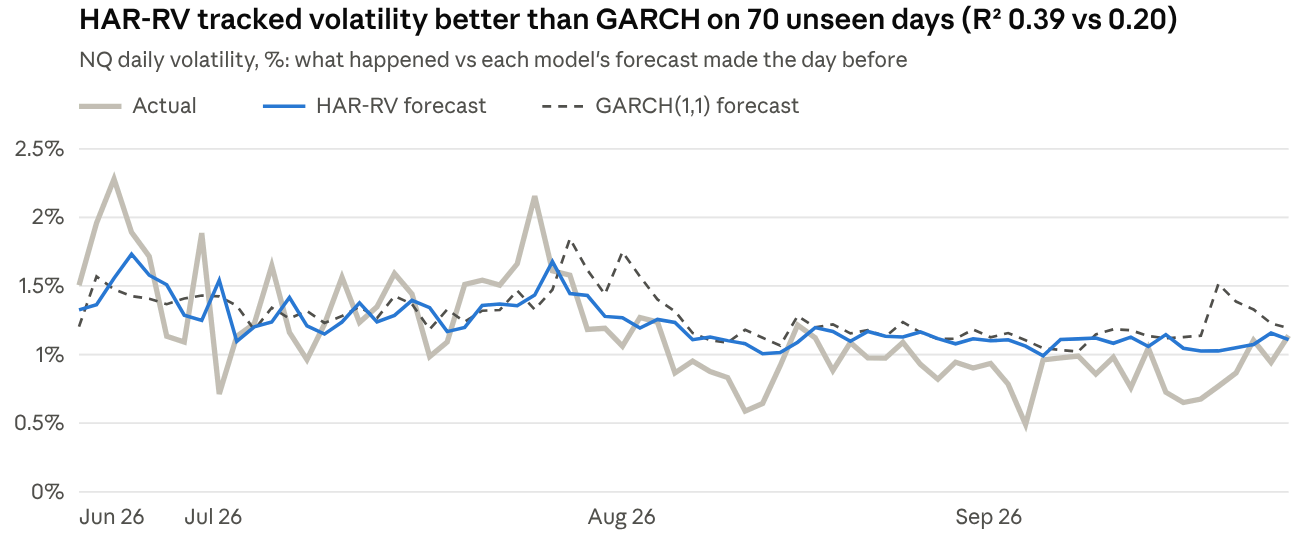
Why the week and month terms were weak. 162 days is short for a model meant to capture monthly memory. On many years of data, as in Corsi's original paper, all three terms are usually significant. Keep them anyway: they cost nothing, and they matter when a calm or wild regime lasts weeks.

Why the t-stats use HAC (Newey-West) errors. The weekly and monthly averages overlap from one day to the next, so the residuals are autocorrelated. Ordinary t-stats would then look too good; HAC errors correct for that. This is the same correction you met for autocorrelated residuals in your course.

In-sample R^2 was 0.17. That sounds low, but daily volatility has a large random part. What counts is how it does on days it has not seen (Step 5).

Step 5: Testing it

On 70 days the model never saw, HAR-RV's forecasts came closest to what really happened.



your differenced_data.csv · test days 23 Jun – 28 Sep 2026; models fitted only on 1 Oct 2025 – 19 Jun 2026

Model	R^2 (higher = better)	MSE (lower = better)	QLIKE (lower = better)
HAR-RV	0.39	0.62	0.121
Yesterday's RV as the forecast	0.32	0.69	0.135
GARCH(1,1) with fat tails, daily returns	0.20	0.81	0.151

How the test was fair. Each model was fitted once on the first 162 days and then frozen. Every forecast used only data up to the day before. "Yesterday's RV" is the simplest rival: tomorrow = today. A model that cannot beat it is not worth using.

What QLIKE is. The standard score for volatility forecasts. It punishes forecasting too low more than forecasting too high, which matches the risk: being caught by a wild day you called calm costs more.

Why GARCH lost. Daily GARCH sees one return per day. HAR-RV's input is built from about 270 five-minute moves per day, so it simply has far more information about each day.

What the chart shows. HAR-RV follows the wild weeks of late June and July closely. Both models over-predict the calm stretch from mid-August, but GARCH more so, with big overshoots in early August and late September. Neither model catches single surprise days: those come from news, not from past volatility.

HAR-RV vs GARCH vs ARIMA

Use ARIMA for the center line, GARCH for bar-by-bar volatility in a live loop, and HAR-RV when you have intraday data and want the next day's (or next hour's) volatility.

	ARIMA	GARCH(1,1)	HAR-RV
Forecasts	the price (the mean)	volatility	volatility
Input	one price series	returns	realized volatility, built from intraday data

	ARIMA	GARCH(1,1)	HAR-RV
Fitted by	maximum likelihood	maximum likelihood	OLS (plain regression)
Can fail to converge	yes	yes	no
Long memory	no	only a slow fade	approximately yes
Uses intraday detail for a daily forecast	no	no	yes
Best use	the forecast line	the range in your live loops	next-day or next-hour risk, stops and position size

They combine well: ARIMA for the line, HAR-RV for the width of the range, the same way your ARIMA notebook already uses GARCH for the width.

Python recipe

This builds daily HAR-RV from MT5 5-minute bars and prints the next day's volatility. I tested it on your NQ data through a stand-in for MT5; it printed a next-day volatility of 1.14%.

```
import numpy as np, pandas as pd, statsmodels.api as sm
import MetaTrader5 as mt5

SYMBOL = "XAUUSDm"

# 1) 5-minute closed bars from MT5 (about 250 trading days)
mt5.initialize()
rates = mt5.copy_rates_from_pos(SYMBOL, mt5.TIMEFRAME_M5, 1, 70000)
mt5.shutdown()
bars = pd.DataFrame(rates)
close = bars.set_index(pd.to_datetime(bars["time"], unit="s"))["close"]

# 2) daily realized variance in %^2 = sum of squared 5-minute log returns
r = np.log(close).diff()
r = r[close.index.to_series().diff() <= pd.Timedelta("30min")]   # drop jumps
over weekend/break gaps
day = r.index.floor("D")
rv = (r.pow(2) * 1e4).groupby(day).sum()
rv = rv[r.groupby(day).count() >= 200]                           # keep full
```

```
trading days only

# 3) the three HAR parts, and tomorrow's value as the target
X = pd.DataFrame({"d": rv, "w": rv.rolling(5).mean(), "m":
rv.rolling(22).mean()})
data = X.assign(y=rv.shift(-1)).dropna()
fit = sm.OLS(data["y"], sm.add_constant(data[["d", "w",
"m"]])).fit(cov_type="HAC", cov_kwds={"maxlags": 5})
print(fit.params.round(3))

# 4) forecast for the next day, from today's three parts
now = X.dropna().iloc[-1]
rv_next = fit.params["const"] + (fit.params[["d", "w", "m"]] * now).sum()
print(f"Next day's volatility: {np.sqrt(rv_next):.2f}% (1 sigma)")
```

Things to check when you run it on gold:

- **History:** if MT5 returns far fewer than 70,000 bars, your broker's history is shorter; you need at least about 100 full days for a stable fit.
- **Day boundary:** days are cut at the broker's midnight, which is fine for gold and BTC.
- **For a next-hour forecast** (closer to your live loops): build hourly RV from 1-minute returns and use the last hour, the last day (about 23 hours) and the last week as the three parts. The code is the same with different windows.

Going further

Once plain HAR-RV works, these are the usual upgrades, roughly in the order people try them:

- **Log-HAR:** model log(RV) instead of RV. Forecasts can never go negative, and single spike days pull the fit less.
- **HAR-J and HAR-CJ:** split each day's RV into smooth movement and sudden jumps (news, data releases), and give each its own weight.
- **HARQ:** gives less weight to days whose RV was measured with more noise.
- **Intraday HAR:** hourly parts (last hour, last day, last week) for next-hour forecasts.
- **Realized GARCH:** a GARCH model that takes RV as an extra input, a bridge between your current GARCH notebooks and HAR.

Check yourself

Try each question before reading the answer under it.

1. Why is realized volatility a better measure of one day's volatility than the day's single return?
   - It uses every 5-minute move in the day (about 270 for NQ), not one close-to-close number, so it is far less noisy.
2. What do the "w" and "m" in HAR-RV stand for, and how many days does each average?
   - Weekly, 5 trading days; monthly, 22 trading days.
3. Your fit gives βd = 0.36. In plain words?
   - About 36% of yesterday's realized variance carries into tomorrow's forecast.
4. Why use HAC (Newey–West) standard errors here?
   - The weekly and monthly averages overlap day to day, so residuals are autocorrelated and ordinary t-stats would look too significant.
5. Why can HAR-RV beat GARCH(1,1) on daily forecasts?
   - GARCH sees one return per day; HAR-RV is built from all the intraday moves, and its three averages copy volatility's long memory.
6. Yesterday was a calm day but the past month was wild. What does HAR-RV predict?
   - Higher than yesterday alone suggests, because the weekly and monthly terms pull the forecast up (when their weights are positive).

Sources

- Corsi, F. (2009), "A Simple Approximate Long-Memory Model of Realized Volatility", Journal of Financial Econometrics 7(2), 174–196 — the paper that introduced HAR-RV.
- **Data used in this lesson:** your file differenced_data.csv, NQ 1-minute price changes from 29 Sep 2025 to 29 Sep 2026. The price level was rebuilt from those changes starting at an assumed 24,600, so the % figures are approximate; the comparisons between models are not affected. Days are UTC calendar days with at least 200 five-minute bars.